

Influence of blood flow occlusion on muscle oxygenation characteristics and the parameters of the power-duration relationship

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¹Department of Kinesiology, Kansas State University, Manhattan, Kansas; ²Department of Anatomy and Physiology, Kansas State University, Manhattan, Kansas; and ³Department of Health and Exercise Science, University of Oklahoma, Norman, Oklahoma

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Broxterman RM, Ade CJ, Craig JC, Wilcox SL, Schlup SJ, Barstow TJ. Influence of blood flow occlusion on muscle oxygenation characteristics and the parameters of the power-duration relationship. *J Appl Physiol* 118: 880–889, 2015. First published February 6, 2015; doi:10.1152/jappphysiol.00875.2014.—It was previously (Monod H, Scherrer J. *Ergonomics* 8: 329–338, 1965) postulated that blood flow occlusion during exercise would reduce critical power (CP) to 0 Watts (W), while not altering the curvature constant (W'). We empirically assessed the influence of blood flow occlusion on CP, W' , and muscle oxygenation characteristics. Ten healthy men (age: 24.8 ± 2.6 yr; height: 180 ± 5 cm; weight: 84.6 ± 10.1 kg) completed four constant-power handgrip exercise tests during both control blood flow (control) and blood flow occlusion (occlusion) for the determination of the power-duration relationship. Occlusion CP (-0.7 ± 0.4 W) was significantly ($P < 0.001$) lower than control CP (4.1 ± 0.7 W) and significantly ($P < 0.001$) lower than 0 W. Occlusion W' (808 ± 155 J) was significantly ($P < 0.001$) different from control W' (558 ± 129 J), and all 10 subjects demonstrated an increased occlusion W' with a mean increase of $\sim 49\%$. The present findings support the aerobic nature of CP. The findings also demonstrate that the amount of work that can be performed above CP is constant for a given condition, but can vary across conditions. Moreover, this amount of work that can be performed above CP does not appear to be the determinant of W' , but rather a consequence of the depletion of intramuscular energy stores and/or the accumulation of fatigue-inducing metabolites, which limit exercise tolerance and determine W' .

critical power; curvature constant; oxygen delivery; muscle ischemia

THE ROBUST NATURE OF THE power-duration relationship (and its equivalents for other modes of exercise) has been well established (26, 27, 52). Nevertheless, the precise physiological mechanisms of the curvature constant (W'), and to a lesser degree critical power (CP), have remained elusive. The growing body of evidence supports that CP represents the highest attainable steady state for aerobic energy production without continually drawing on W' and, as such, demarcates the boundary between the heavy- and severe-intensity exercise domains (4, 12, 38, 39, 42, 51). It is also evident that W' is a constant term that determines the limit of exercise tolerance (T_{lim}) for severe-intensity exercise (21, 55). Intramuscular energy stores (35, 36, 38), the accumulation of fatigue-inducing metabolites (7, 18, 21, 28), and/or the magnitude of the severe-intensity domain (5, 51) have all been postulated to determine W' . Building evidence supports that complete utilization of W' is associated with consistent muscle phosphocreatine ([PCr]),

inorganic phosphate ([P_i]), and hydrogen ion concentration ($[H^+]$) perturbations, which may limit the amount of work performed above CP (28, 43, 51). Additionally, the rate of W' utilization (but not the magnitude of W') is influenced by manipulations in O_2 delivery and O_2 extraction via alterations in muscle contraction duty cycle (4). While much has been revealed regarding the mechanisms of CP and W' , it is not clear how blood flow occlusion influences each parameter.

In originally characterizing the power-duration relationship, Monod and Scherrer (38) speculated on the influence of blood flow occlusion by stating, “Factor b [CP] is linked to circulatory conditions in the muscle. For when the dynamic work is performed under arterial cuff, the maximum work becomes constant whatever be the time after which exhaustion occurs. The maximum work is then equal to factor a [W'] . . .” Implicit in this statement is that CP must be 0 Watts (W) with blood flow occlusion for the maximum work performed to be equal to W' and W' to remain constant. This suggests that CP is independent of anaerobic energy production, and that the O_2 trapped in the arm with occlusion (i.e., bound to myoglobin and within the vasculature) is not sufficient to measurably contribute to CP. Furthermore, this statement implies that W' is not determined by aerobic energy production or the clearance of H^+ from the muscle vascular space with perfusion (at least beyond the conditions of the occluded limb). A constant W' also implies that intracellular pathways for energy production are unaltered, if indeed similar intramuscular metabolic perturbations do occur for the complete utilization of W' across interventions. Thus blood flow occlusion exercise is a unique model to empirically test the statement of Monod and Scherrer (38), while providing further insight into the mechanisms determining the power-duration relationship.

Exercise across work rates within the severe-intensity domain (42) is characterized by the concomitant progressive increases in oxygen uptake ($\dot{V}O_2$) and blood flow, the depletion of intramuscular [PCr], and the accumulation of [P_i] and [H^+], all of which achieve similar values at the T_{lim} (3, 9, 28, 32, 38, 43, 49, 51). Moreover, Vanhatalo et al. (51) demonstrated that [PCr], [P_i], and [H^+] at T_{lim} during knee-extension exercise are independent of augmented O_2 delivery via inspired hyperoxic gas. Consistent with these muscle physiological responses being independent of O_2 delivery, similar skeletal muscle electromyography (EMG) characteristics are expressed between normoxic and hypoxic exercise (15, 41). In contrast, fractional O_2 extraction {indicated by deoxygenated hemoglobin + myoglobin concentration, (deoxy[Hb+Mb]) (10, 14, 19, 20, 22)} at T_{lim} appears to be scaled to O_2 delivery, such that it is greater with hypoxia (41) and less with hyperoxia (51) than normoxia, respectively. However, it remains to be determined

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whether similar EMG characteristics and fractional O_2 extractions are attained at T_{lim} for severe-intensity domain power outputs within a given O_2 delivery. Thus it remains to be determined whether reductions in O_2 delivery via blood flow occlusion (rather than inspired hypoxic gas) result in similar EMG characteristics and attainment of fractional O_2 extraction.

Therefore, the aim of the present study was to assess the influence of blood flow occlusion on the parameters of the power-duration relationship (CP and W'), EMG characteristics, and muscle oxygenation characteristics. According to the prediction of Monod and Scherrer (38), our primary hypotheses were that, with blood flow occlusion, 1) CP would not be significantly different from 0 W, and 2) W' would not be significantly different from control. Second, we hypothesized that, within each condition, 3) deoxy[Hb+Mb] and 4) EMG characteristics would not be significantly different at T_{lim} .

METHODS

Experimental procedures. All experimental procedures in the present study were approved by the Institutional Review Board at Kansas State University and conformed to the standards set forth by the Declaration of Helsinki. Before providing written informed consent and completion of a health history questionnaire, subjects were informed of the protocol and potential risks of participation. Testing sessions were separated by at least 24 h, and subjects were instructed to abstain from vigorous activity during the 24 h before testing, in addition to abstaining from caffeine and alcohol consumption during the 2 and 12 h before testing, respectively.

A previously described custom-built two-handed handgrip ergometer (4) was utilized for all testing sessions. The ergometer was attached to a pneumatic cylinder by means of a cable-pulley system,

which provided a fixed linear displacement of 4 cm per handgrip contraction. Resistance was controlled via pressurization of the pneumatic cylinder, and work was accomplished by compressing the air within the pneumatic cylinder. Power output was calculated as $P = Rdfk^{-1}$, where P is power in W, R is resistance in kg, d is displacement in meters (m), f is contraction frequency, and k is the constant 6.12 for the conversion of $kg \cdot m \cdot min^{-1}$ to W. Alterations in power output were accomplished via alterations in resistance, as d and f were held constant. The ergometer was calibrated before the study. Subjects were seated in front of the ergometer and grasped the handrail such that both forearms were approximately at heart level. Exercise was performed using a 50% contraction duty cycle (1.5 s contraction/1.5 s relaxation) at a rate of 20 contractions/min. An audio recording with the specific timing was used in conjunction with feedback provided by an investigator to ensure correct timing. Before data collection, subjects completed three testing sessions for familiarization with the contraction protocol. All testing sessions were continued until T_{lim} , determined as the inability to successfully complete three consecutive contraction cycles.

An incremental power output test was first performed for the determination of peak power (P_{peak}). This protocol was initiated at 1.0 W, and the power output was increased by 0.5 W/min until T_{lim} . The greatest power output for which at least one-half of the stage was completed was used as P_{peak} . Each subject also completed four constant-power tests for both the control brachial artery blood flow (control) and occluded brachial artery blood flow (occlusion) conditions in a randomized order. The P_{peak} was utilized to determine power outputs that would elicit T_{lim} within ~1–15 min (43). The control constant-power tests were performed at 80, 90, 110, and 130% P_{peak} , and the occlusion constant-power tests were performed at ~17 (1 W for all subjects), 35, 80, and 110% P_{peak} . Pilot work in our laboratory demonstrated these intensities elicited similar T_{lim} values for occlusion constant-power tests as those recommended for control

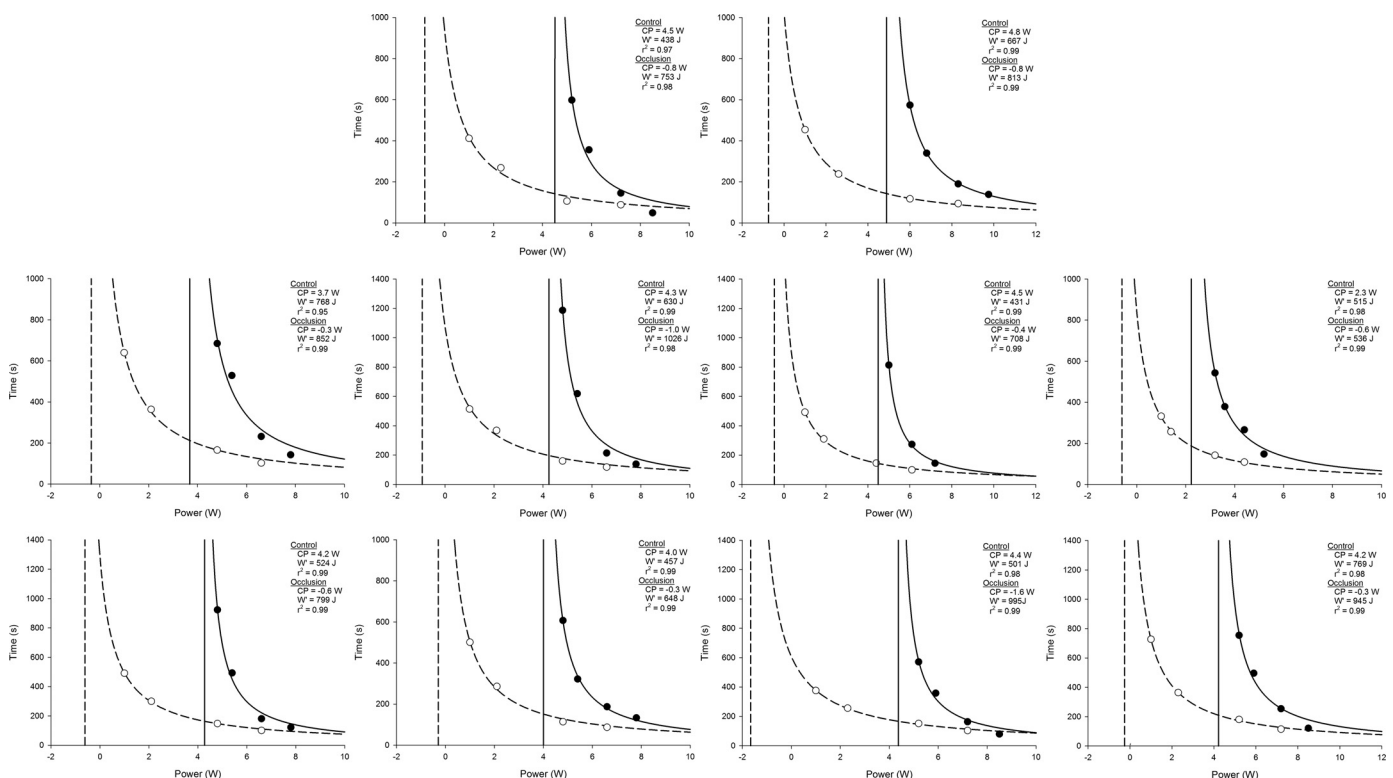


Fig. 1. Individual subject constant-power and hyperbolic curve fit data. Subjects completed four constant-power tests for both control and occlusion at power outputs selected to elicit exhaustion in ~1–15 min. Each subject's constant-power data were fit with a two-parameter hyperbolic model to provide parameter estimates for critical power (CP) and the curvature constant (W').

constant-power tests. The 80 and 110% P_{peak} were conducted in both conditions to examine the influence of blood flow, independent of differences in power output. Brachial artery blood flow was occluded with a vascular cuff positioned around the brachial region of each arm, which was rapidly inflated (<0.3 s) to suprasystolic pressures (≥ 275 mmHg) at the onset of exercise and remained inflated until T_{lim} (E20 Rapid Cuff Inflator, Hokanson, Bellevue, WA). Blood flow occlusion was verified by the absence of a radial pulse, and the cuff pressures were continuously monitored to ensure ≥ 275 mmHg.

Measurements and data analysis. The power-duration relationship for both control and occlusion was determined by fitting the power output and T_{lim} data from the constant-power tests with the two-parameter hyperbolic model:

$$t = W' / (P - CP)$$

where t is T_{lim} in s, W' is the finite work capacity in Joules (J), P is the power output in W, and CP is in W.

A frequency-domain multidistance near-infrared spectroscopy (NIRS) system was used to measure the oxygenation characteristics of the flexor digitorum superficialis of the right forearm during each testing session (Oxiplex TS, ISS, Champaign, IL). Detailed descriptions of the principles and algorithms of the NIRS technology have previously been described (20, 23). Briefly, this device consists of one detector fiber bundle and eight light-emitting diodes (LED) operating at wavelengths of 690 and 830 nm (four LEDs per wavelength). The LED-detector fiber bundle separation distances are 2.0, 2.5, 3.0, and 3.5 cm. This NIRS device measures and incorporates the dynamic reduced scattering coefficients to provide absolute concentrations (μM) for deoxy[Hb+Mb], oxygenated [Hb+Mb] (oxy[Hb+Mb]), total-[Hb+Mb], and %saturation [Hb+Mb] (%sat-[Hb+Mb]). The deoxy[Hb+Mb] is relatively insensitive to changes in blood volume (10, 19, 22) and has been used to reliably estimate the fractional oxygen extraction (10, 13, 14, 19, 20, 22, 34). The NIRS probe was calibrated before each test, according to the manufacturer's recommendations. The belly of the flexor digitorum superficialis of the right arm was identified using palpation and EMG. The NIRS probe was secured along the belly of the flexor digitorum superficialis and was wrapped with an elastic bandage to prevent movement of the probe. The position of the NIRS probe was marked with indelible ink for reproducible placement throughout the study. The NIRS data were collected at 50 Hz and stored for post hoc analysis. The NIRS data were analyzed using 1-s time-binned mean values and 9-s time-binned mean values at T_{lim} . A kinetics analysis was conducted for the deoxy[Hb+Mb] data over the initial 60 s of exercise using a mono-exponential model:

$$\text{deoxy} - [\text{Hb} + \text{Mb}](t) = \text{deoxy} + [\text{Hb} + \text{Mb}](b) + A(1 - e^{-(t - \text{TD})/\tau})$$

where deoxy[Hb+Mb](t) is the deoxy[Hb+Mb] at any point in time, deoxy[Hb+Mb](b) is the baseline deoxy[Hb+Mb] before the onset of exercise, A is the amplitude of the deoxy[Hb+Mb] response, TD is the time delay before the start of the increase in deoxy[Hb+Mb], and τ is the time constant of the increase in deoxy[Hb+Mb].

Surface EMG (Trigno EMG, Delsys, Boston, MA) measurements were obtained from the flexor digitorum superficialis of the left forearm during each testing session. The belly of the left flexor digitorum superficialis was identified by palpation and strong electrical activity when the fingers were flexed, but not with ulnar or radial deviation. The position of the EMG sensor was marked with indelible ink for reproducible placement throughout the study. The EMG data were collected at 1,000 Hz and stored for post hoc analysis. The raw EMG data were processed with a band-pass filter (0.05–400 Hz), and each electrical burst corresponding to a muscle contraction was detected using a custom-designed computer program. For each muscle contraction, the signal amplitude characteristics were analyzed via root mean square (RMS) to provide an index of muscle activation and

motoneuron firing rate. RMS values were normalized to the first minute value of each test. The frequency characteristics were analyzed via median power frequency (MedPF) to provide an index of the muscle action potential conduction velocity.

Data analysis. All curve-fitting procedures and statistical analyses were performed using a commercially available software package (SigmaPlot and SigmaStat, Systat Software, Point Richmond, CA). Student's paired t -tests were used to compare the CP and W' parameters measured for control and occlusion. Occlusion CP was compared with 0 W using a one-sample t -test. One-way ANOVAs with repeated measures were used to compare NIRS or EMG measurements within each condition. Two-way ANOVAs with repeated measures (condition $\times$ intensity) were used to compare both the NIRS and EMG measurements for the tests performed at 80 and 110% P_{peak} . Tukey's post hoc analyses were conducted when significant main effects were detected, and effect sizes (ES) were calculated for both CP and W' . Differences were considered statistically significant when $P < 0.05$. 95% confidence intervals (CI) around CP and W' were determined for each subject in both conditions. All data are presented as means $\pm$ SD, unless otherwise noted.

RESULTS

Ten healthy men (age: 24.8 ± 2.6 yr; height: 180 ± 5 cm; weight: 84.6 ± 10.1 kg) completed the study. The P_{peak} from the incremental ramp test was 6.1 ± 0.9 W. The durations of the constant-power tests were as follows: 80% P_{peak} , 759 ± 243 ; 90% P_{peak} , 471 ± 156 ; 110% P_{peak} , 211 ± 45 ; and 130% P_{peak} , 122 ± 32 s, for control; and 17% P_{peak} , 494 ± 118 ; 35%

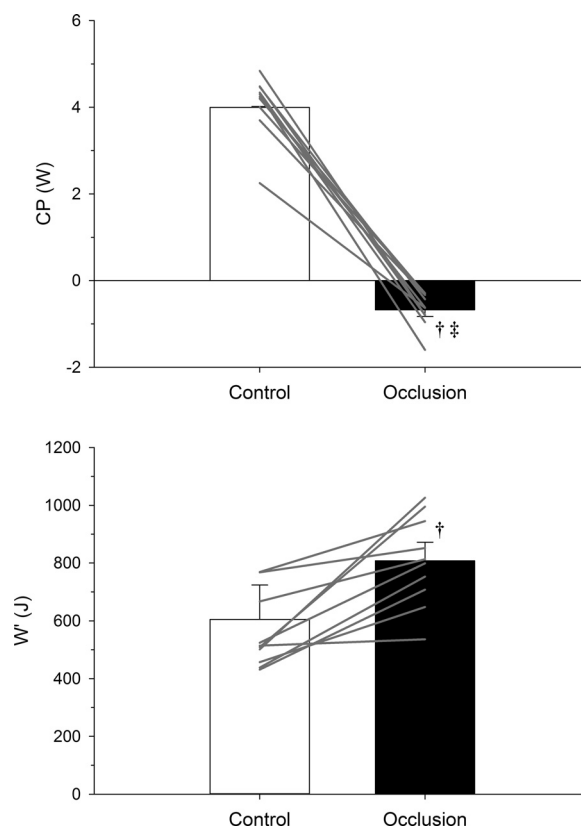


Fig. 2. Mean CP and W' determined for control and occlusion handgrip exercise. CP and W' were determined for both control blood flow (control) and brachial artery blood flow occlusion (occlusion). The shaded lines indicate individual subject responses. Values are means $\pm$ SE. †Significantly different from control ($P < 0.001$). ‡Significantly different from 0 W ($P < 0.0001$).

P_{peak} , 301 ± 49 ; 80% P_{peak} , 143 ± 24 ; and 110% P_{peak} , 102 ± 10 s, for occlusion. The time to achieve T_{lim} was significantly shorter for occlusion compared with control at both 80% P_{peak} (143 ± 24 s vs. 759 ± 243 s, $P < 0.002$) and 110% P_{peak} (102 ± 10 s vs. 211 ± 45 s, $P < 0.001$).

Individual subject hyperbolic model fits are presented in Fig. 1. The determination of the control power-duration relationship in one subject was found to be greatly weighted toward a specific constant-power test, as previously described (4, 12, 38, 39, 42, 51). Therefore, this subject's control power-duration relationship was determined using three constant-power tests (4, 12, 38, 39, 42, 51). The mean hyperbolic model fits (r^2) were 0.98 ± 0.02 for control and 0.99 ± 0.01 for occlusion. Occlusion CP (-0.7 ± 0.4 W; mean 95% CI = -1.3 – 0.0 W) was significantly ($P < 0.001$, ES = 6.0) lower than control CP (4.1 ± 0.7 W; mean 95% CI = 3.4 – 4.7 W) and significantly ($P < 0.001$, ES = 1.75) lower than 0 W (Fig. 2). Occlusion W' (808 ± 155 J; mean 95% CI = 533 – $1,082$ J) was significantly ($P < 0.001$, ES = 1.4) different from control W' (558 ± 129 J; mean 95% CI = 158 – 982 J), and all 10 subjects demonstrated an increased occlusion W' with a mean increase of $\sim 49\%$ (Fig. 2).

The NIRS measurements were not significantly different at T_{lim} within each blood flow condition across the different intensities, except for %sat-[Hb+Mb] between 35 and 110% within occlusion (Figs. 3 and 4; Table 1). Total-[Hb+Mb] increased significantly above baseline for all control exercise

tests, and the values at T_{lim} were not significantly different among these tests. In contrast, total-[Hb+Mb] did not significantly increase above baseline during the occlusion tests (Fig. 4), demonstrating complete blood flow occlusion of the limb. For the 80 and 110% P_{peak} tests, deoxy[Hb+Mb] at T_{lim} was significantly greater for occlusion than control, and oxy[Hb+Mb], total-[Hb+Mb], and %sat-[Hb+Mb] were significantly lower for occlusion than control (Fig. 5; Table 1). The deoxy[Hb+Mb] kinetics analyses results are presented in Table 2. There were no significant differences in the TD or τ between control and occlusion, while the A was significantly greater for occlusion than control at both 80 and 110% P_{peak} .

No statistically significant differences for control RMS at T_{lim} were detected, while the MedPF at T_{lim} was significantly less for 110% P_{peak} compared with both 80% P_{peak} ($P < 0.001$) and 90% P_{peak} ($P = 0.005$). Occlusion RMS at T_{lim} was significantly greater for 110% P_{peak} compared with 17% P_{peak} ($P < 0.001$). There were no significant differences in occlusion MedPF at T_{lim} (Fig. 6). RMS at T_{lim} was significantly lower for occlusion compared with control at 110% P_{peak} ($P = 0.012$), but there was no significant difference at 80% P_{peak} between occlusion and control ($P = 0.106$). Occlusion MedPF at T_{lim} was significantly lower than control at 80% P_{peak} ($P < 0.001$), and there was no significant difference at 110% P_{peak} between occlusion and control ($P = 0.43$).

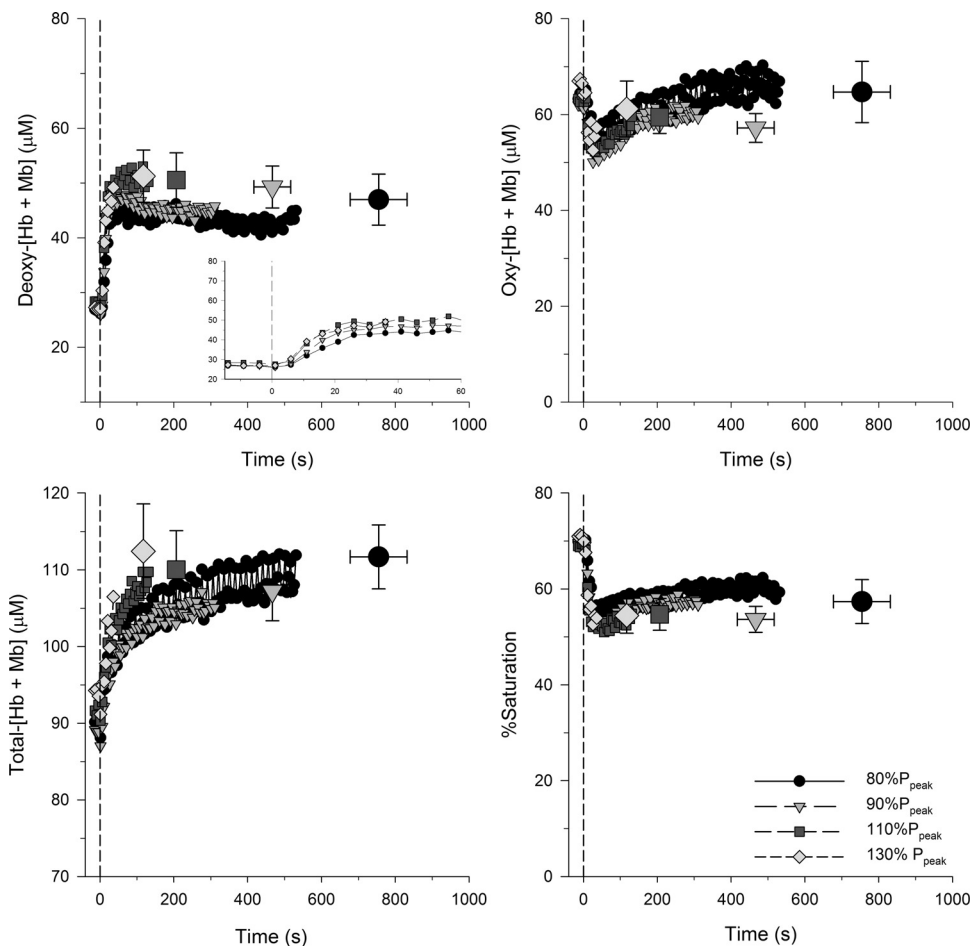
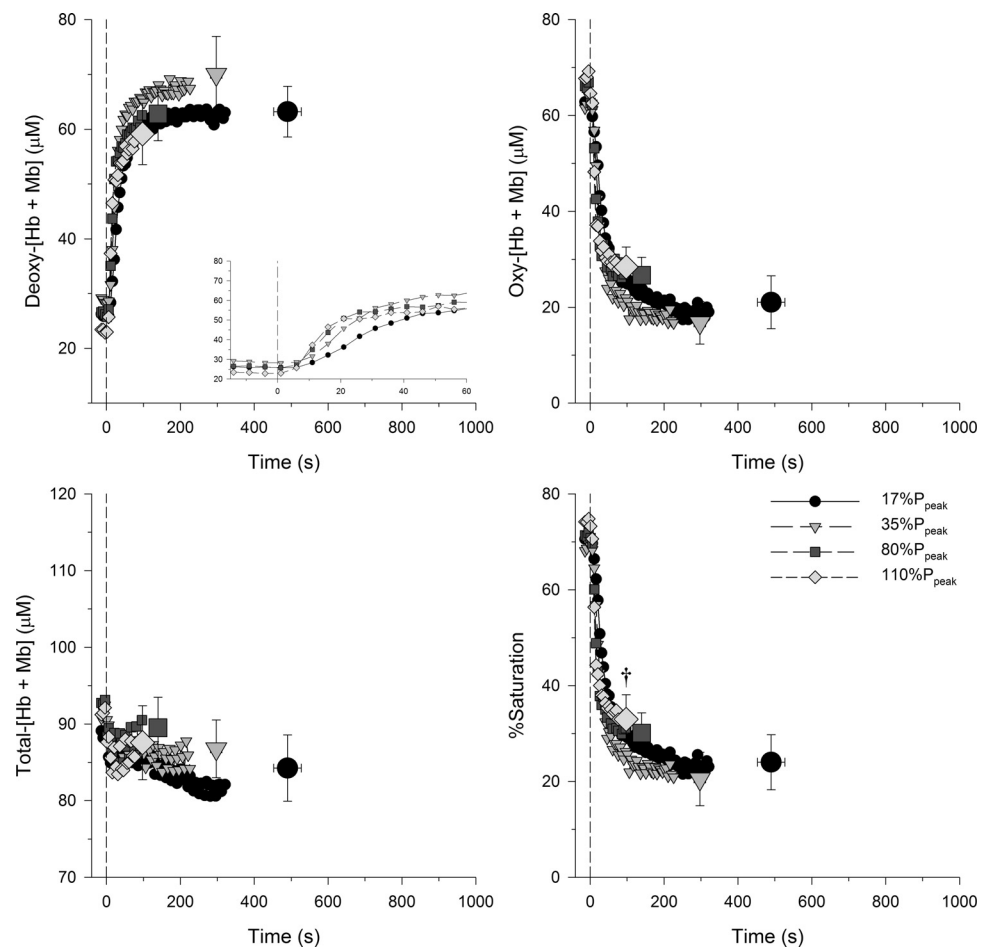


Fig. 3. Mean near-infrared spectroscopy (NIRS) muscle oxygenation data for control exercise. Shown are deoxygenated (deoxy [Hb+Mb]), oxygenated (oxy[Hb+Mb]), total (total-[Hb+Mb]), and percent saturation of total hemoglobin + myoglobin concentration (%sat-[Hb+Mb]) data during 80, 90, 110, and 130% peak power (P_{peak}). Values are means $\pm$ SD. No significant differences were detected between intensities at the limit of exercise tolerance (T_{lim}). Graph inset: deoxy[Hb+Mb] during the initial 60 s of exercise onset.

Fig. 4. Mean NIRS muscle oxygenation data for occlusion exercise. Deoxy[Hb+Mb], oxy[Hb+Mb], total-[Hb+Mb], and %sat-[Hb+Mb] data during 17, 35, 80, and 110% P_{peak} are shown. Values are means $\pm$ SD. †Significantly ($P < 0.05$) different from 110% P_{peak} at T_{lim} . Total-[Hb+Mb] did not significantly increase above baseline for occlusion. Graph inset: deoxy[Hb+Mb] during the initial 60 s of exercise onset.



DISCUSSION

This study examined the influence of blood flow occlusion on the parameters of the power-duration relationship during handgrip exercise. Handgrip exercise during blood flow occlusion was well-described by the two-parameter hyperbolic model. In contrast to the primary hypotheses, occlusion CP was lower than 0 W, and occlusion W' was greater than control W' . These results support the aerobic nature of CP, as the reduction in O_2 delivery with occlusion reduced CP. These results also support that W' is relatively constant within a given O_2 delivery condition, but can vary across O_2 delivery conditions. Additionally, this study presents novel findings regarding muscle oxygenation characteristics during control and occluded severe-intensity handgrip exercise. In agreement with our third hypothesis, deoxy[Hb+Mb] at T_{lim} was similar within each condition. In contrast to our fourth hypothesis, differences in EMG characteristics at T_{lim} were detected within each condition.

The power-duration relationship. A two-parameter hyperbolic model may be used to empirically describe the decrease in T_{lim} with increasing power outputs to yield the parameters CP and W' (38, 43, 55). The asymptote in this model represents CP, which distinguishes an exercise intensity above which a physiological steady state is not attained and exercise T_{lim} is limited by W' . As such, CP represents the highest sustainable rate of aerobic ATP production for which W' will

not be continuously utilized (4, 12, 38, 39, 42, 51). The curvature constant in this model represents W' , modeled as a finite amount of work that can be performed above CP that, when fully expended, results in T_{lim} (21, 38, 43, 55). However, mounting evidence suggests that W' is not solely a finite amount of work, per se. Rather it appears that similar “limit-

Table 1. End-exercise NIRS values for control and occlusion

P_{peak} , %	Deoxy[Hb+Mb], μM	Oxy[Hb+Mb], μM	Total-[Hb+Mb], μM	Sat-[Hb+Mb], %
<i>Control</i>				
80	47.0 $\pm$ 14.7	64.7 $\pm$ 20.2	111.70 $\pm$ 13.2	57.4 $\pm$ 14.5
90	50.0 $\pm$ 12.1	57.2 $\pm$ 9.5	107.2 $\pm$ 12.2	53.7 $\pm$ 8.6
110	50.6 $\pm$ 15.7	59.5 $\pm$ 10.7	110.0 $\pm$ 16.1	54.7 $\pm$ 10.5
130	51.3 $\pm$ 15.1	31.2 $\pm$ 18.4	112.4 $\pm$ 19.5	54.5 $\pm$ 11.7
<i>Occlusion</i>				
17	63.9 $\pm$ 14.6	21.0 $\pm$ 17.4	84.2 $\pm$ 13.7	24.0 $\pm$ 18.2
35	70.0 $\pm$ 21.9	16.8 $\pm$ 14.0	86.8 $\pm$ 11.9	20.5 $\pm$ 17.7*
80	62.8 $\pm$ 15.5†	26.7 $\pm$ 11.8†	89.5 $\pm$ 12.6†	30.1 $\pm$ 13.3†
110	59.2 $\pm$ 17.7†	28.4 $\pm$ 13.2†	87.6 $\pm$ 15.2†	33.0 $\pm$ 16.2†

Values are means $\pm$ SD. P_{peak} , peak power from the incremental maximal exercise test; deoxy[Hb+Mb], oxy[Hb+Mb], total-[Hb+Mb], and Sat-[Hb+Mb]: deoxygenated, oxygenated, total and saturation of hemoglobin + myoglobin concentration, respectively. †Significantly ($P < 0.05$) different from control at the same % P_{peak} . *Significantly ($P < 0.05$) different from 110% P_{peak} within condition.

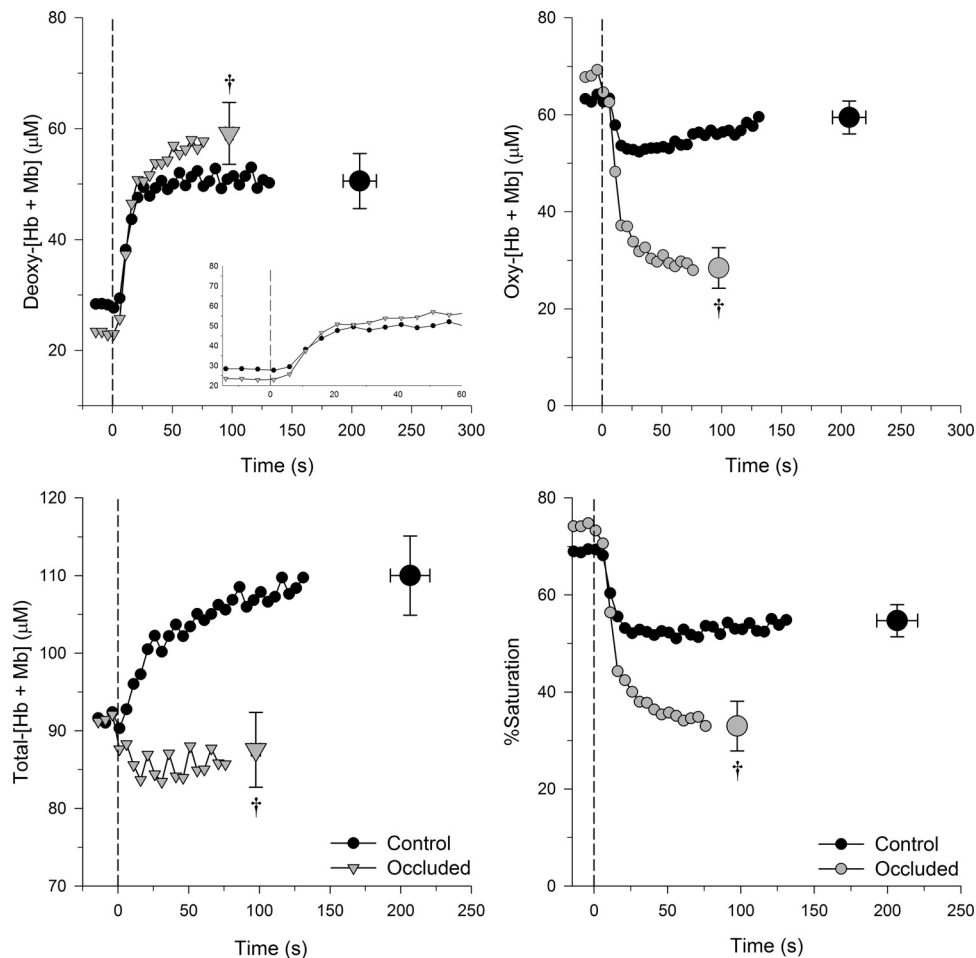


Fig. 5. Mean NIRS muscle oxygenation data for control and occlusion at 110% P_{peak} . Deoxy[Hb + Mb], oxy[Hb + Mb], total-[Hb + Mb], and %sat-[Hb + Mb] at 110% P_{peak} are shown. Values are means $\pm$ SD. †Significantly ($P < 0.05$) different from control. Total-[Hb + Mb] did not significantly increase above baseline for occlusion. Graph inset: deoxy[Hb + Mb] during the initial 60 s of exercise onset.

ing" muscle metabolic perturbations in [PCr], $[P_i]$, and $[H^+]$ are attained, that in turn constrain the maximal amount of work that can be performed above CP (6, 28, 43, 51). Despite the precise mechanisms of CP and W' not being fully understood, it is clear that exercise intensities above CP are predictably

limited by W' , and that the T_{lim} occurs at similar physiological states for these intensities.

Deterministic mechanisms of CP. To date, evidence suggests that CP represents the highest attainable steady state for aerobic energy production without continuously drawing upon W' (4, 12, 38, 39, 42, 51). It was with this interpretation of CP in mind that Monod and Scherrer (38) postulated that the maximum amount of work performed under blood flow occlusion would be equal to W' , which would necessitate a CP equal to 0 W. In contrast to this prediction, the present study demonstrated that the calculated occlusion CP was actually reduced to a power output slightly, but significantly, <0 W. While a negative CP is only theoretical and not physiologically attainable, nonetheless, this estimate provides mechanistic insight into CP. An occlusion CP equal to 0 W (i.e., rest) would indicate that indefinite resting occlusion would not hinder the ability of the skeletal muscle to perform contractions. Previous studies have demonstrated hemoglobin and myoglobin deoxygenation and the subsequent depletion of [PCr] during resting occlusion (25, 31). Thus, after sufficient depletion of aerobic energy sources, resting metabolism is sustained by anaerobic energy production, and the depletion of [PCr] and accumulation of $[P_i]$ suggest that W' is utilized during resting occlusion (i.e., 0 W). As CP is the highest sustainable rate of aerobic ATP production without drawing continuously upon W' , it would not be expected that occlusion CP be equal to 0 W, as, without

Table 2. Exercise onset deoxy[Hb + Mb] kinetics parameters for control and occlusion

P_{peak} , %	y_0 , μ M	A, μ M	τ , s	TD, s
<i>Control</i>				
80	26.7 ± 4.8	17.8 ± 8.4	10.2 ± 3.4	$8.8 \pm 3.0^*$
90	26.7 ± 5.3	21.8 ± 11.5	11.6 ± 12.2	6.9 ± 1.5
110	28.1 ± 6.7	22.1 ± 11.2	7.8 ± 2.6	6.7 ± 1.8
130	27.0 ± 5.7	24.0 ± 11.6	12.0 ± 11.6	4.5 ± 2.4
<i>Occlusion</i>				
17	26.1 ± 5.4	38.1 ± 15.7	32.2 ± 14.3	11.2 ± 3.1
35	$28.7 \pm 7.0^\S$	38.0 ± 17.4	$17.4 \pm 7.1^\ddagger$	$10.4 \pm 2.7^\S$
80	27.6 ± 5.5	$33.4 \pm 10.1^\dagger$	$8.7 \pm 2.6^\ddagger$	$8.4 \pm 2.0^\ddagger^\S$
110	23.0 ± 4.6	$34.0 \pm 16.0^\dagger$	$12.1 \pm 9.9^\ddagger$	$4.9 \pm 3.6^\ddagger$

Values are means $\pm$ SD. Data were analyzed over the first 60 s of exercise with a monoexponential model. y_0 , Baseline concentration; A, amplitude of response; τ , time constant of response; TD, time delay from exercise onset to the increase in response. Significantly ($P < 0.05$) different from †control, *130% P_{peak} within condition, ‡17% P_{peak} within condition, and §110% P_{peak} within condition.

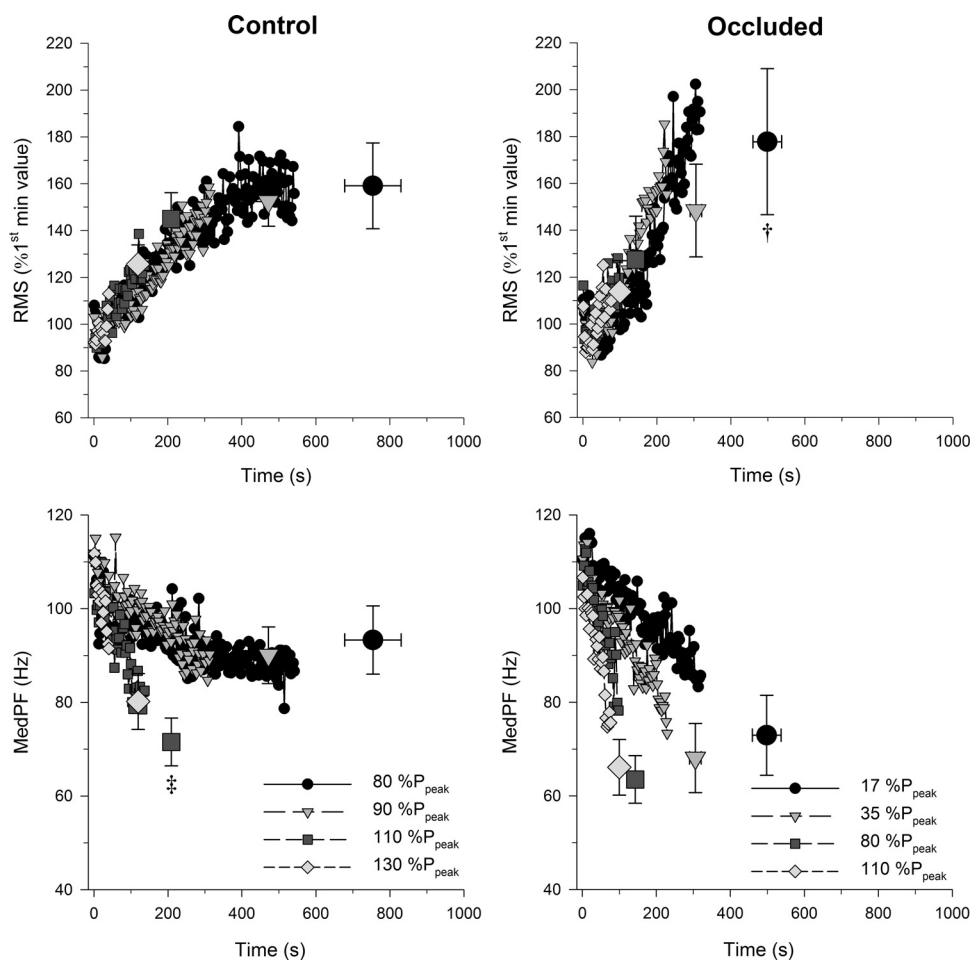


Fig. 6. Mean EMG data for control and occlusion. Root mean square (RMS) and median power frequency (MedPF) data at all intensities for control and occlusion are shown. Values are means $\pm$ SD. †Significantly different from 17% P_{peak} . ‡Significantly different from 80 and 90% P_{peak} within control.

blood flow, there is no sustainable rate of aerobic ATP production. Rather, CP would be expected to be a theoretical value of negative power for which the magnitude is proportional to the resting metabolic rate.

Deterministic mechanisms of W' . W' was significantly greater with blood flow occlusion in all 10 subjects (mean increase $\sim 49\%$). A greater W' with occlusion may be due to differences in metabolic economy between occlusion and control exercise (30, 31) altering the degree of metabolic perturbation for a given amount of mechanical work. Moreover, as T_{lim} during occlusion appears to be the result of the accumulation of inhibitory by-products of glycolytic ATP turnover (31), it may be that a greater degree of accumulation is tolerated during occlusion exercise and thus more work above CP may be performed. Furthermore, if the 250 J increase in W' for occlusion arose solely from oxidative energy production, it would require a greater O_2 extraction of 12 ml (assuming 0.000239 kcal/J and 5 kcal/l O_2). The greater deoxy[Hb+Mb] for occlusion than for control suggests that a small portion of the increase in W' may be due to an increased O_2 extraction. However, the greater deoxy[Hb+Mb] for occlusion ($\sim 64 \mu M$) than for control ($\sim 50 \mu M$) would amount only to 0.05 ml O_2 (assuming 150 g flexor digitorum superficialis and 1,060 g/l muscle density), which represents 0.4% of the 12 ml O_2 needed for the increase in W' to be the result of O_2 extraction. Moreover, no relationship between the increase in deoxy[Hb+Mb] and W' was detected. Thus it does not appear that

the increased W' during occlusion is the result of the greater fractional O_2 extraction.

The results of the present study lend credence to W' not being determined by a finite amount of work per se, but rather by other mechanisms such as the rate of attainment of "limiting" muscle metabolic perturbations (7, 21, 28, 43, 51), the magnitude of the severe domain (5, 51), and/or the $\dot{V}O_2$ slow component (40). Modeling the present data reveals that, while W' is utilized in its entirety for exercise intensities above CP, the proportion of W' that contributes directly to external work is not constant across this range of power outputs (Fig. 7). For example, at rest (i.e., 0 W) under occlusion, W' will be used in its entirety to support factors distinct from external work (i.e., resting cellular processes, ion handling, etc.). This is demonstrated by resting blood flow occlusion leading to a desaturation in myoglobin (46), increased adenosine diphosphate, and a decreased [PCr] (2, 25). With increasing power outputs, a greater proportion of W' would be utilized for external work, as the proportion of energy turnover via external work will increase (Fig. 7). Thus, at sufficiently high power outputs (i.e., Fig. 7, bottom right), the majority of W' is associated with external work. However, it appears that some of the energy derived from the utilization of W' still contributes to the factors that are distinct from external work, including the internal work of handgrip contraction. This model may explain why W' (determined as the amount of external work performed above CP) is constant when determined under normal blood flow

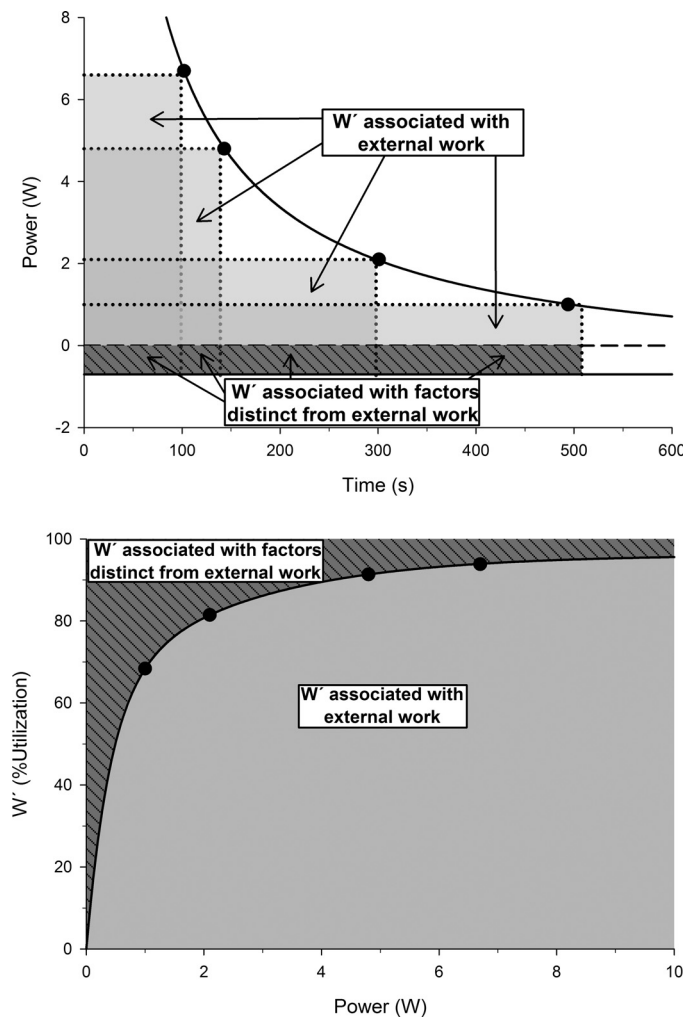


Fig. 7. Schematic diagram demonstrating the contribution of W' to mechanical- and nonmechanical-work energy consuming processes. *Top*: CP (the horizontal asymptote) and W' for blood flow occlusion handgrip exercise. CP was significantly below 0 W. The area defined by the power output and the duration of exercise between CP and 0 W represents the amount of W' associated with factors distinct from external work (i.e., ion handling, resting metabolic rate, internal work, etc.), and the area between 0 W and power output is assumed to represent the amount of W' associated with external work. *Bottom*: the mean amount of W' associated with external work and factors distinct from external work (i.e., ion handling, resting metabolic rate, internal work, etc.) at the four occlusion exercise intensities that accounted for the amount of W' positioned between CP and 0 W. This demonstrates the nature of W' and that varying proportions of W' are utilized for different processes, depending on the intensity of the exercise.

conditions, as power outputs utilized to determine the normal power-duration relationship are on the *top right* portion of the curve in Fig. 7, *bottom*. This is consistent with previous suggestions that W' is determined by an integration of multiple mechanisms, rather than a single mechanism alone (7, 17).

Muscle oxygenation characteristics. In the present study, fractional O_2 extraction was significantly greater for occlusion than for control. This is consistent with previous reports of alterations in the fractional O_2 extraction with varying inspired O_2 concentrations (41, 46, 51). These alterations in fractional O_2 extraction may be the consequence of the Bohr effect. In the present study, an elevated $[H^+]$ with occlusion (31) may have

facilitated oxyhemoglobin dissociation, resulting in the greater fractional O_2 extraction (50).

Fick's law of diffusion states that the flux of O_2 ($\dot{V}_{O_2}$) is dependent on the diffusivity of oxygen ($\dot{D}_{O_2}$) and the PO_2 gradient between the microvasculature (PO_{2mv}) and the mitochondria (PO_{2mit}): $\dot{V}_{O_2} = \dot{D}_{O_2} \times (PO_{2mv} - PO_{2mit})$. Exercise hyperemia increases microvascular hematocrit and, therefore, enhances $\dot{D}_{O_2}$ (16, 24, 29, 33). The similar peak plateaus in microvascular hematocrit (i.e., total-[Hb+Mb]) in the present study for control exercise suggest that the peak microvascular hematocrit, and, therefore, $\dot{D}_{O_2}$, is constrained for a given condition. The peak $\dot{D}_{O_2}$ gradient ($PO_{2mv} - PO_{2mit}$) during severe exercise would also be constrained, as intramuscular PO_2 (PO_{2mit}) achieves similar low values during exercise (37, 44–46) (assuming similar PO_{2mv} at T_{lim} for a given condition). These constraints directly impact aerobic energy production, thus dictating maximum $\dot{V}_{O_2}$ (44, 53, 54) and presumably CP (4, 12, 39, 51). CP in the present study was greatly attenuated during occlusion, due presumably, in part, to the prevented increase in microvascular hematocrit (i.e., no increase total-[Hb+Mb] above baseline) and the expected exacerbated fall in PO_{2mv} . As such, W' would be utilized earlier and at a faster rate in the occlusion exercise bouts compared with control, thus resulting in the earlier attainment of T_{lim} . Moreover, the similar nadirs above zero in oxy[Hb+Mb] at T_{lim} for occlusion exercise suggest that complete deoxygenation of Hb and/or Mb did not occur, consistent with previous findings that Hb and Mb do not fully desaturate during blood flow occlusion exercise (11, 31, 47, 48). These similar nadir values of deoxygenation may result from the continual fall in PO_{2mv} (decrease in oxy[Hb+Mb]) as oxygen is utilized and not replenished during occlusion exercise, until the capillary-mitochondria PO_2 gradient achieves equilibrium with the inherent resistance to $\dot{D}_{O_2}$ ($1/\dot{D}_{O_2}$). At this point, further O_2 flux into the mitochondria would be prevented (i.e., oxy[Hb+Mb] nadir). Additionally, the accumulation of metabolic by-products (i.e., $[H^+]$) may inhibit oxidative phosphorylation (8), despite the presence of oxygen remaining in the microcirculation (i.e., oxy[Hb+Mb] above 0 μM). Therefore, severe-intensity exercise tolerance appears to be dependent on perfusive and diffusive O_2 supply dictating CP, along with the magnitude of W' and its rate of utilization (i.e., the power output).

During occlusion exercise, the Fick principle [$\dot{V}_{O_2} = \dot{Q} \times (a-v)O_2 \text{ diff}$], where $\dot{Q}$ is cardiac output and $(a-v)O_2 \text{ diff}$ is arteriovenous O_2 difference, dictates that $\dot{V}_{O_2}$ is proportional to O_2 extraction. Interestingly, the τ for the change in the deoxy[Hb+Mb] was not significantly different between conditions at the same power output, while the A was greater for occlusion. Consistent with this, Vanhatalo et al. (51) demonstrated an unchanged τ for deoxy[Hb+Mb] between hyperoxic and normoxic knee-extension exercise. These results suggest that O_2 delivery, per se, does not determine the τ for deoxy[Hb+Mb]. Rather, it appears that, at the onset of exercise, sufficient levels of O_2 are present within the muscle and surrounding tissue so as to not limit O_2 flux, despite O_2 delivery interventions. It is not until later (approximately >30 s) into the exercise bout that the O_2 delivery interventions begin to exert influence on fractional O_2 extraction.

Motor unit recruitment and firing characteristics. In the present study, RMS and MedPF at T_{lim} were not always similar within each condition. Moreover, significant differences were

detected between control and occlusion RMS at 80% P_{peak} and MedPF at 110% P_{peak} . These findings differ from those of prior studies, demonstrating similar EMG characteristics between normoxia and hypoxia (15, 41). The present results may differ from previous findings due to greater magnitude of O_2 delivery reduction with occlusion compared with hypoxia. Together, these results suggest that EMG characteristics appear not to be influenced by reductions in O_2 delivery up to a certain point, but further reductions in O_2 delivery past this point result in EMG characteristic differences.

Implications of current findings. The findings of the present study further demonstrate the integration of perfusive and diffusive O_2 delivery in determining aerobic energy production and, therefore, CP. It is this established CP that dictates the intensities at which W' will be continually utilized until fatigue ensues. Specifically, the present findings indicate that reductions in blood flow (i.e., O_2 delivery) lower CP, which results in the utilization of W' and fatigue at lower intensities. Cumulatively, current evidence suggests that alteration in O_2 delivery via changes in blood flow or inspired O_2 concentrations directly affects CP and W' utilization (4, 12, 39, 51).

Limitations. Several experimental limitations must be considered when interpreting the present data. It is not known from the present data if similar metabolic perturbations are occurring within the muscle during occlusion and control. Therefore, it remains to be determined if the same limiting muscle metabolic perturbations are being achieved at T_{lim} during blood flow occlusion exercise as during control. With this, the present study and many previous studies have utilized small muscle mass exercise (e.g., handgrip and knee extension). It remains to be determined whether large muscle mass exercise (i.e., whole body exercise, such as cycling and running) elicits similar intramuscular metabolic responses for power outputs within the severe-intensity domain. Furthermore, the present study is not able to determine which other nonmechanical work energy consuming processes may be utilizing W' . Previous publications have demonstrated that a substantial amount of energy is utilized for ion pumping during muscular contraction (1). However, it is not known if these previous findings are applicable in the extreme case of blood flow occlusion exercise. As such, it currently is not known how the relationship between W' and the severe-intensity domain is altered with blood flow occlusion. As [PCr], $[P_i]$, and $[H^+]$ were not measured in the present study, it cannot be certain that the consistent muscle oxygenation coincided with consistent concentration of these intramuscular metabolites. Prior evidence suggests that consistent values for these metabolites are to be expected, despite alterations in O_2 delivery (28, 49, 51). However, it remains to be determined if these values are altered with blood flow occlusion. The noise inherent in EMG measurements limits the confidence in conclusively interpreting the data. As a result, the EMG data are not presented to precisely state the EMG characteristics of the muscle, but rather to suggest that the muscle may not be achieving similar EMG characteristics at T_{lim} within and between O_2 delivery conditions. Future studies are needed to more completely evaluate the EMG responses for each condition.

Conclusions. The present study demonstrated a greater fractional O_2 extraction during occlusion exercise compared with control. Moreover, muscle oxygenation characteristics attained similar values within a given O_2 delivery condition. Addition-

ally, EMG characteristics appear to be independent of O_2 delivery, but rather to be determined by the performed power output. Cumulatively, current evidence suggests that intramuscular [PCr], $[P_i]$, and $[H^+]$ at T_{lim} are similar and independent of alterations in O_2 delivery, but that fractional O_2 extraction is directly influenced by alterations in O_2 delivery. Moreover, the reduction of O_2 delivery with blood flow occlusion exercise decreased the estimated CP below 0 W and led to an increased W' compared with control. It appears that the resting metabolic rate and/or the internal work of handgrip exercise may result in the estimated apparent CP being below 0 W during occlusion. The findings of the present study support the aerobic nature of CP. Additionally, the findings demonstrate that the amount of work that can be performed above CP is constant for a given condition, but can vary across conditions. Moreover, this amount of work that can be performed above CP does not appear to be the determining mechanism of W' , but rather a consequence of the depletion of intramuscular energy stores and/or the accumulation of fatigue inducing metabolites, which limit exercise tolerance and determine W' .

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DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the author(s).

AUTHOR CONTRIBUTIONS

Author contributions: R.M.B., C.J.A., J.C.C., S.L.W., S.J.S., and T.J.B. conception and design of research; R.M.B., C.J.A., J.C.C., S.L.W., S.J.S., and T.J.B. performed experiments; R.M.B. and T.J.B. analyzed data; R.M.B. and T.J.B. interpreted results of experiments; R.M.B. and T.J.B. prepared figures; R.M.B. and T.J.B. drafted manuscript; R.M.B., C.J.A., J.C.C., S.L.W., S.J.S., and T.J.B. edited and revised manuscript; R.M.B., C.J.A., J.C.C., S.L.W., S.J.S., and T.J.B. approved final version of manuscript.

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